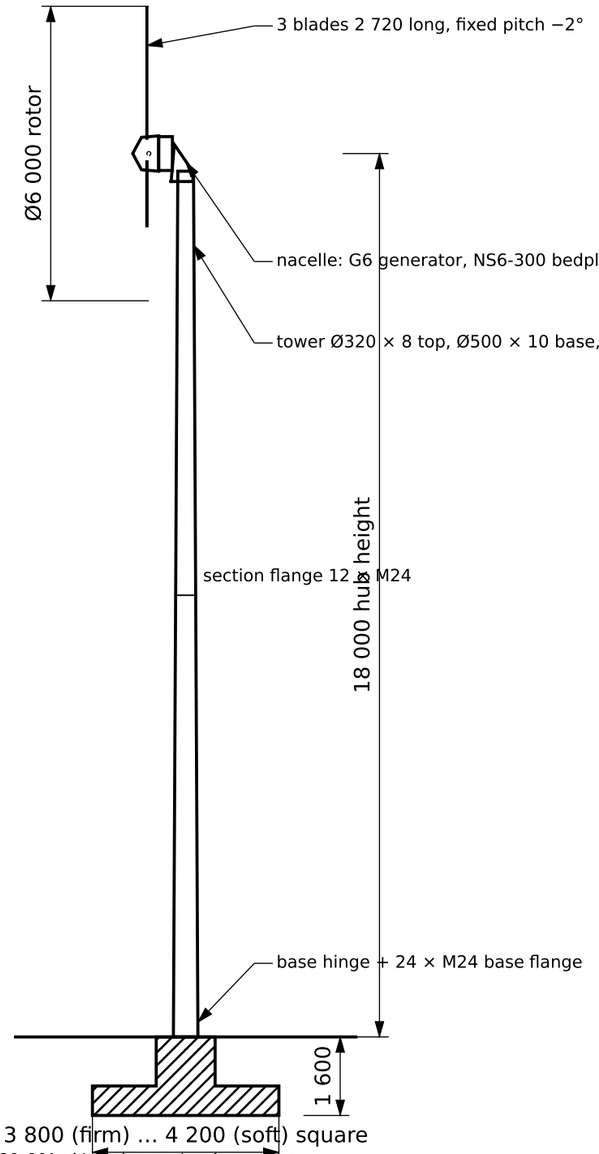
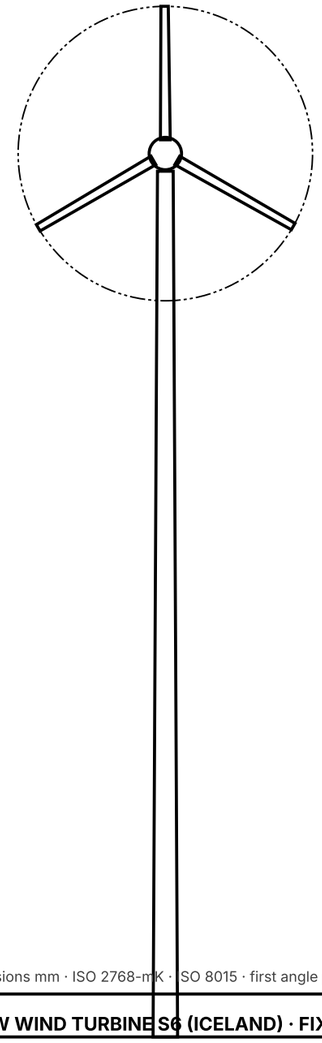
Item	Value
Rated power	6 kW electric at ≈ 9.7 m/s
Rotor	Ø6.0 m, 3 blades, TSR 5.5, fixed pitch −2°
Speed	41–164 rpm, trip 189 rpm
Control	stall by speed (converter), storm ramp 25–30 m/s
Stopping	K1 braking resistor 3 × 4 Ω + spring brake .5 kNm
Storm	stop, yaw 90° out, hold
Hub height	18 m, tilt-up monopole
Tower f1	0.87 Hz (exclusion 47.5–57.4 rpm)
Head mass	≈ 912 kg above the tower flange
Tower mass	1824 kg

SIDE VIEW (wind from the left)



VIEW FROM UPWIND



NOTES

1. Docs 28–32. Shared 12 kW parts are listed on NS6-001 with their own drawing numbers.
2. Rotor plane 750 mm upwind of the yaw axis, shaft tilt 5°, rotor centre 570 mm above the yaw bearing top face.
3. Footing size depends on the ground class (doc 30 §6): firm 3.8 m / 9.5 m³, frost silt 3.9 m / 10.0 m³, soft volcanic 4.2 m / 11.4 m³; rock: 1.6 × 1.6 × 0.8 m plinth with 8 anchors (NS6-500).

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

GENERAL ARRANGEMENT · TURBINE

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-000	A · 2026-11-06	1:125
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see parts list	≈ 912 kg head + 1824 kg tower	Claude / not checked

Item	Drawing	Part	Source	Qty	Mass kg
1	NS6-100	Blade, glass/epoxy, glass UD caps	new	3	20.6
2	NS6-200	Fixed hub weldment S355NL + shims ±2°	new	1	87
3	NS6-210	Spinner (GRP)	new	1	6
4	Rev C gen.	G6 generator: Rev C stator, drum, magnets, no fins/ribs	doc 23/29	1	263
5	NS6-610	Plain rear cover (replaces the bladed Rev C cover)	new	1	7
6	NP1-100/101	Spindle 42CrMo4 Ø140/90 + spacer	12 kW	1	58
7	NP1-230	Brake disc 16 mm	12 kW	1	34.5
8	NP1-310	Brake caliper bracket (one of two positions used)	12 kW	1	–
9	bought	Spring caliper, 1.5 kNm spring pack	doc 29	1	≈ 9
10	NP1-320	Rotor lock pin and holder	12 kW	1	4
11	NS6-300	Bedplate weldment, 6 kW (lighter)	new (doc 32)	1	127
12	NY-400…460	Yaw ring, adapter, 2 gearmotors, pads, lock, encoder, hanger	12 kW doc 25	1 set	≈ 275
13	NS6-400	Tower 18 m, 2 sections, base hinge	new size	1	1824
14	NS6-500	Foundation: spread footing or rock plinth	new size	1	concrete
15	NS6-600	Braking resistor box + K1 + safety chain	new (doc 29)	1	≈ 18
16	doc 12	Converter, dump chopper, 48 V module (6 kW firmware)	12 kW	1	–

NOTES

1. The 12 kW drawings named here (cad/nacelle_p1 NP1-xxx, cad/yaw NY-4xx, generator Rev C sheets in project files generator/revC) are used unchanged.
2. G6 differences from Rev C (doc 29): inner fins, outer ribs and cover blades left off; one caliper with a lighter spring pack; braking resistor and contactor added.
3. NS6-300 replaces the 12 kW bedplate NP1-300 (doc 32). Same interfaces: spindle flange, yaw inner ring, gearmotor brackets NP1-420, pad units NY-410, lock pin holder, cable hanger.
4. Masses: calculated. Head mass above the tower flange in doc 32 §2.

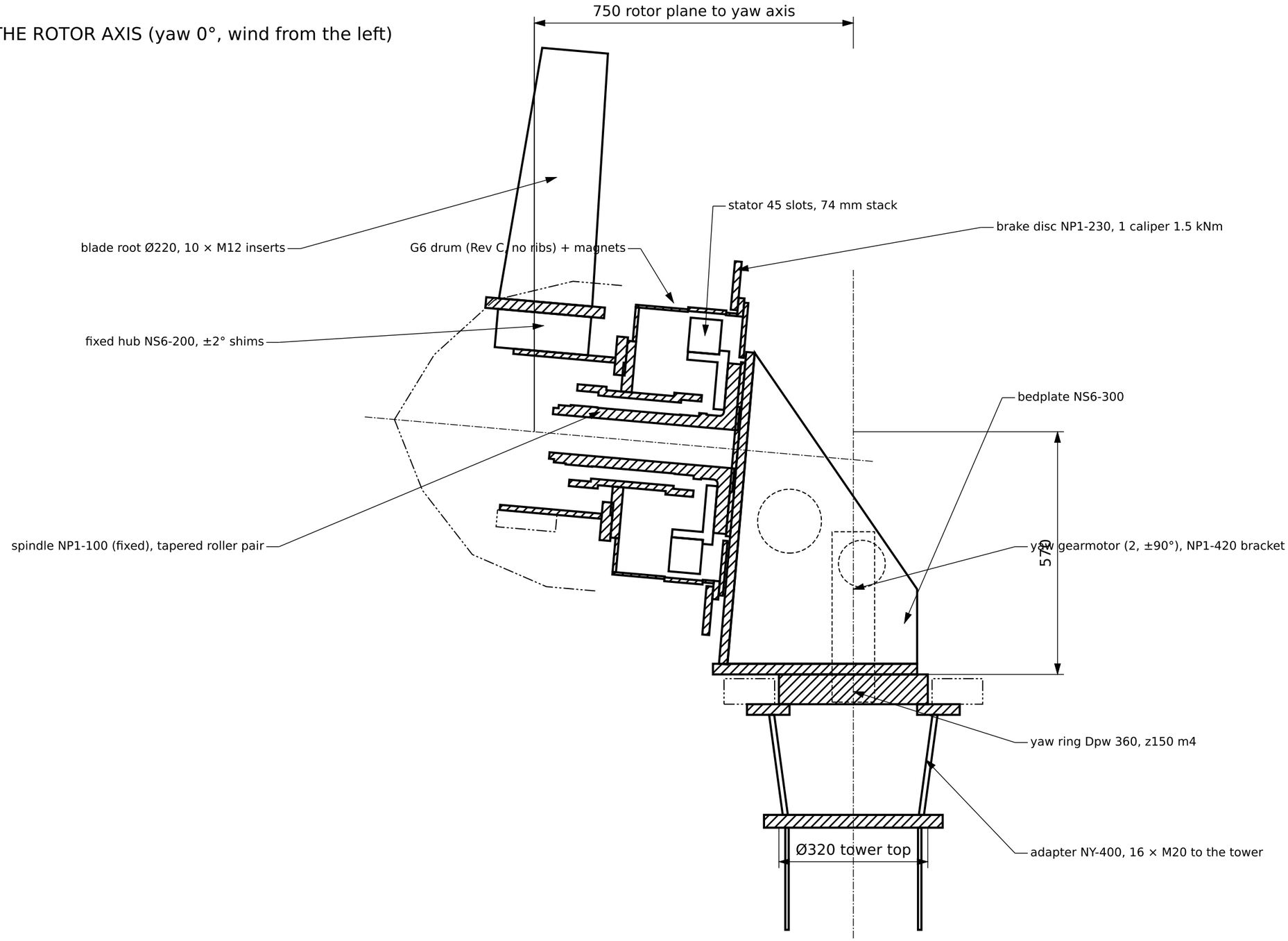
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

PARTS LIST AND REUSED 12 kW DRAWINGS

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-001	A · 2026-11-06	–
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
–	–	Claude / not checked

SECTION ON THE ROTOR AXIS (yaw 0°, wind from the left)



NOTES

1. Drivetrain geometry from cad/nacelle_p1/geometry.py (Rev B nacelle, Rev C generator, doc 23); fins, ribs and cover blades not fitted on G6 (doc 29).
2. Rotor turns with the drum on two tapered roller bearings on the stationary spindle; the hub bolts to the drum front face (12 × M16 PCD 400).
3. Section shows the hub arm of the upper blade; the other two arms are at ±120°. Spinner (phantom) is GRP, NS6-210.
4. Braking resistor box NS6-600 is mounted on the tower base cabinet wall, not in the nacelle (heat). Phase leads run in the hanger tube.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

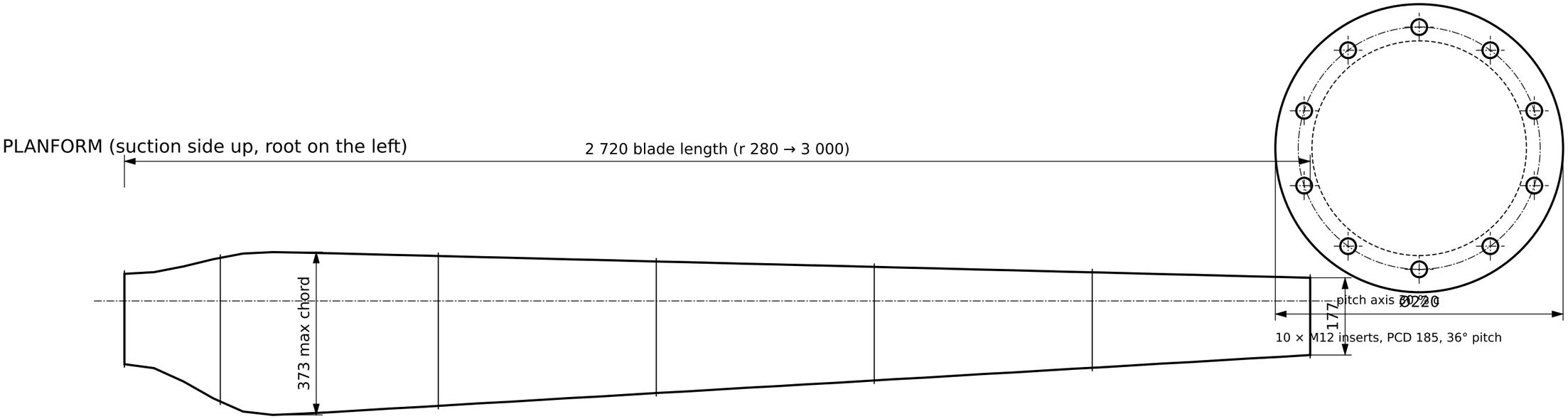
6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

GENERAL ARRANGEMENT · NACELLE SECTION

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-010	A · 2026-11-06	1:10
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see NS6-001	≈ 912 kg above the tower flange	Claude / not checked

ROOT END (from the hub)

PLANFORM (suction side up, root on the left)



r mm	chord	twist °	t/c	cap mm
280	207	20.0	1.00	4.25
400	254	20.0	0.71	4.25
600	370	20.0	0.32	4.25
800	360	17.1	0.28	4.25
1000	343	13.4	0.26	4.10
1400	310	8.5	0.24	3.38
1800	277	5.5	0.21	2.41
2200	244	3.6	0.19	1.44
2600	210	2.2	0.18	1.00
3000	177	1.2	0.18	1.00

NOTES

1. Aerofoil family and twist from analysis/rotor/bem.py (TSR 5.5, cl 1.0, R 3.0 m, root r 0.28 m); twist given at the pitch axis, positive nose up towards feather; blade set at −2° with the hub shims (NS6-200).
2. Structure doc 30 §2 / doc 10 model: two glass UD caps (table), ±45° glass skins, PVC foam core, single shear web 1.5 mm glass; root wall builds up 12 mm over the first 0.7 m. Cap buckling at minimum thickness governs (use 1.00); strength 0.43 (SLM H), fatigue 0.10 with simulated loads (doc 31).
3. Root: 10 bonded steel inserts M12 (IKEA-type sleeve), PCD 185, pull-out ≥ 60 kN each (test). Bolt force 43 kN at SLM H.
4. Mass 20.6 kg ± 0.3 kg; static moment matched within 0.5 % per set of three (balance class doc 09). Erosion tape on the outer 1 m of the leading edge.
5. First flap 15.2 Hz, first edge 38.3 Hz at standstill. Tip deflection 0.06 m, tower clearance 0.83 m.

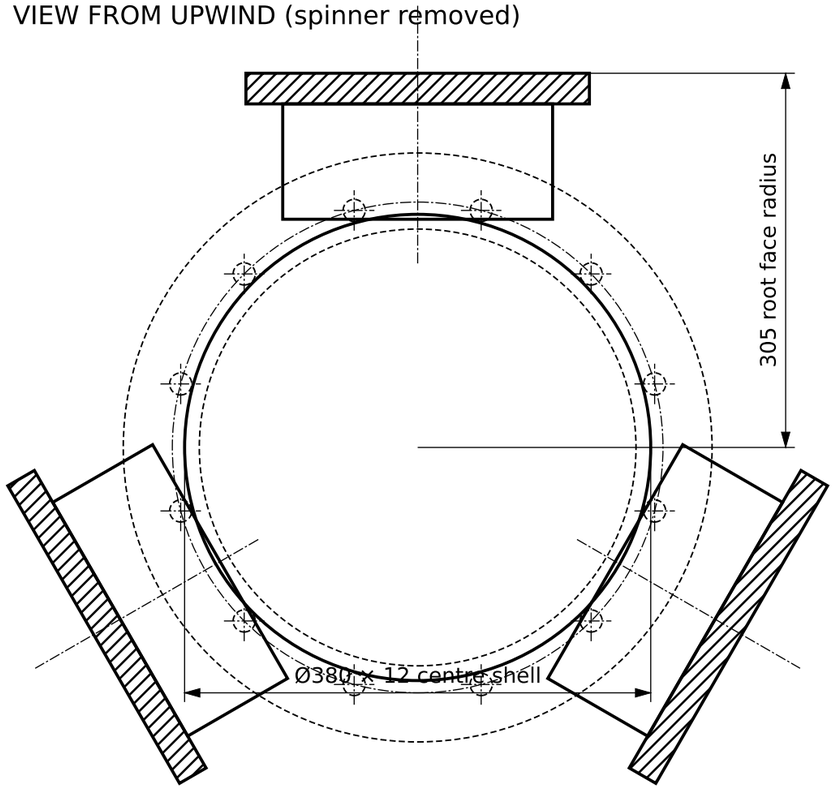
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

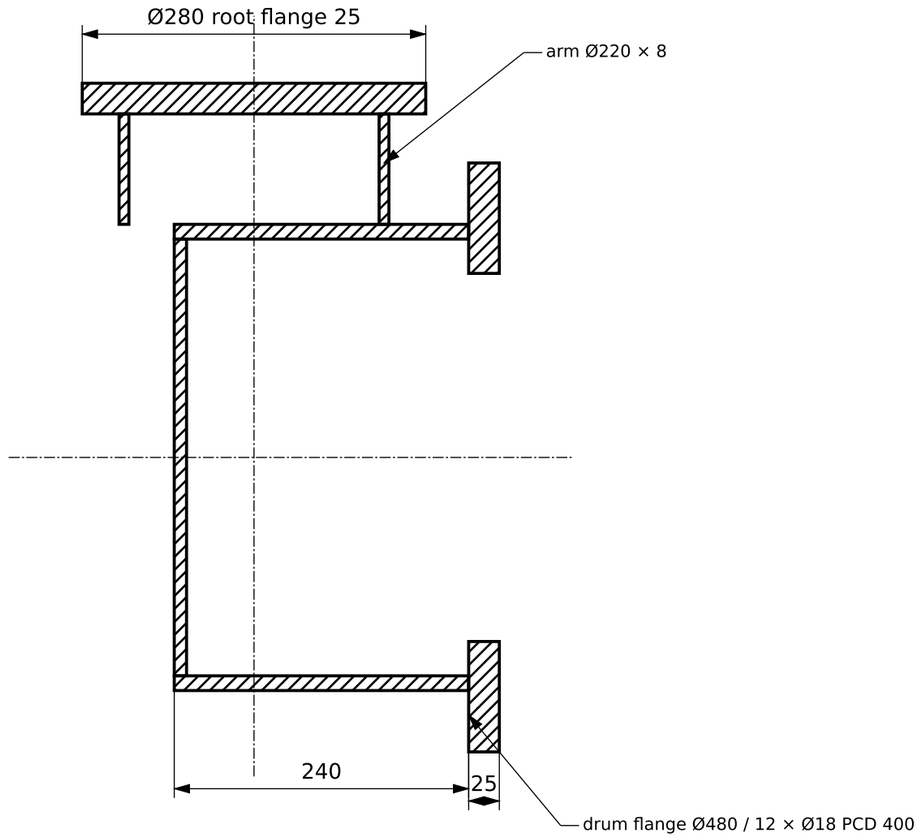
BLADE · PLANFORM, TWIST AND ROOT

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-100	A · 2026-11-06	1:10
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
E-glass/epoxy infused; glass UD caps; PVC core shell	20.6 kg	Claude / not checked

VIEW FROM UPWIND (spinner removed)



SECTION THROUGH THE UPPER ARM



NOTES

1. Arms Ø220 × 8: parked SLM stress 68 MPa (allow 323); fatigue at the flange weld 5.7 MPa with simulated loads (allow 30.8, detail 71).
2. Drum joint: 12 × M16 10.9 PCD 400 (Rev C interface), friction grip, slip OK at 5.7 kNm design torque.
3. Root flange faces machined after stress relief at −2.0° ± 0.05° pitch about each arm axis; 10 × M12 clearance holes Ø13 PCD 185; shim seat for ±2° tapered shims (0.5° steps).
4. Welds full penetration at arm-to-shell and arm-to-flange, ISO 5817 B, 100 % MT + UT on the flange welds. Hot-dip galvanised or zinc sprayed + C5 paint.
5. Front plate Ø356 × 10 closes the shell; spinner brackets (3 × M8) on the arms. FE check of the arm-to-shell weld before release (doc 30 §11).

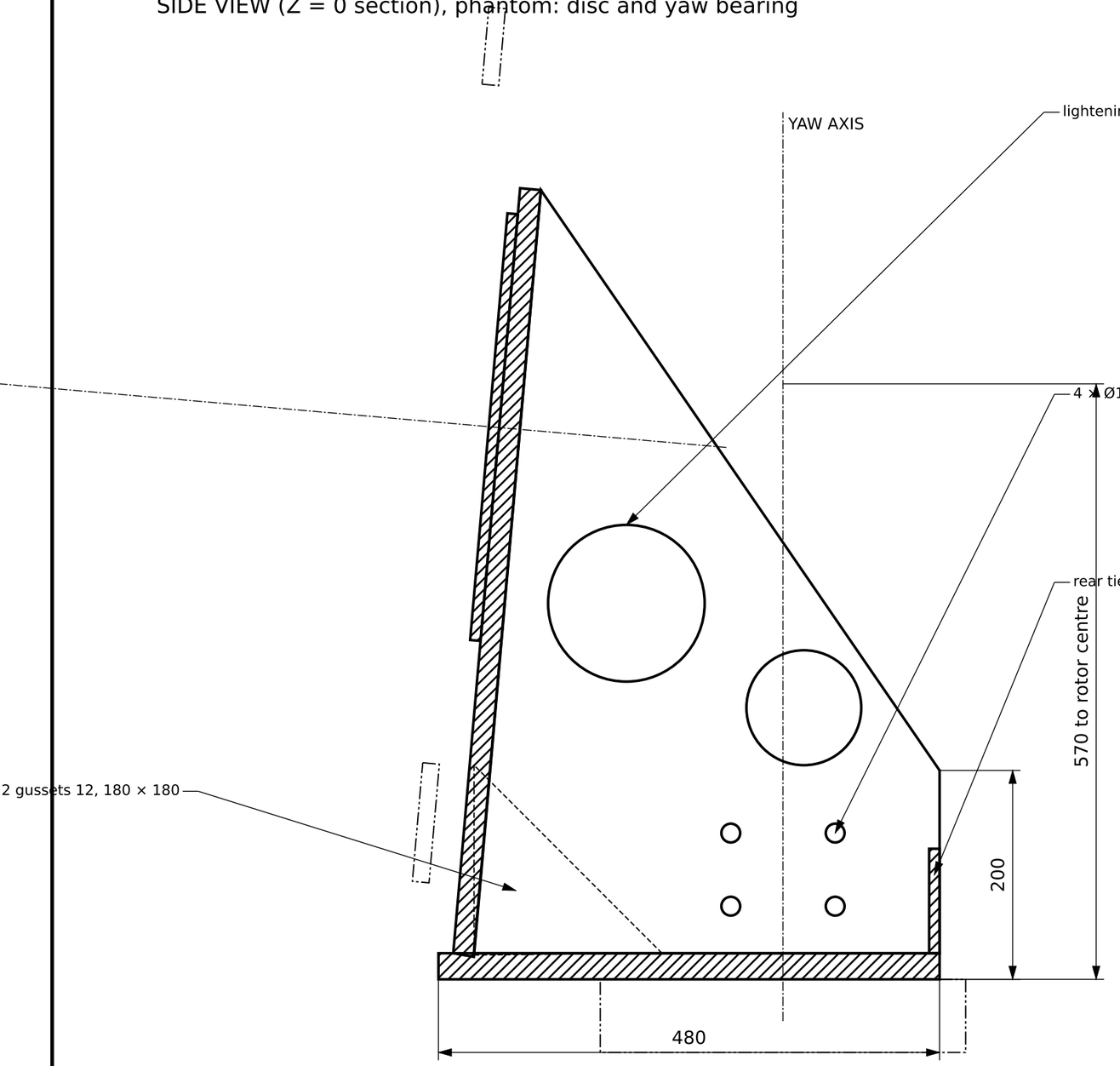
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

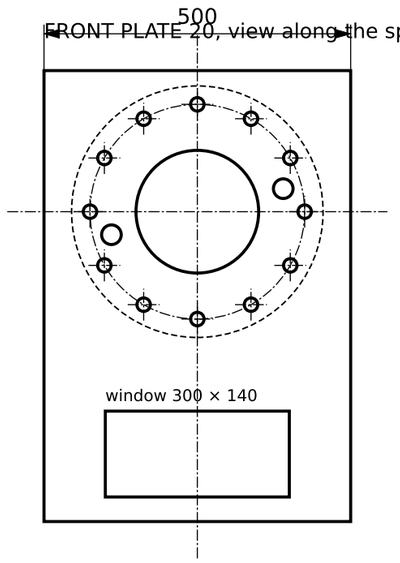
FIXED HUB WELDMENT

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-200	A · 2026-11-06	1:5
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
S355NL (EN 10025-3), welded, stress relieved, machined	87 kg (+ 6 kg spinner)	Claude / not checked

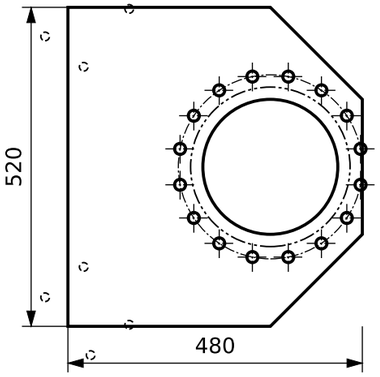
SIDE VIEW (Z = 0 section), phantom: disc and yaw bearing



FRONT PLATE 20, view along the spindle axis



BOTTOM PLATE (yaw flange), from below



NOTES

1. Replaces NP1-300 (12 kW, 333 kg as drawn) on S6 (doc 32). Same interfaces: Ø410 × 10 pad at x 470, Ø200 H7 spigot, 12 × Ø22 PCD 350 (M20 10.9 from inside, 540 Nm), 2 × Ø32 cable holes PCD 290; bottom Ø220 bore, 16 × Ø18 PCD 300 (yaw inner ring, M16 10.9 with nuts).
2. Plates: front 20, bottom 25, sides 10 (outer faces Y ±260 for NP1-420), rear tie 10, gussets 12. Rear corners of the bottom plate chamfered 150 × 150.
3. Pad units NY-410 bolt under the front corners (4 × M12 each, tapped, hidden); hanger tube flange NY-460 on Ø260 at the bore; caliper bracket NP1-310 on the left side plate front edge; lock pin holder NP1-321 on 2 × M12 in the front plate top edge.
4. Design checks (SLM γ 3, doc 32): root section 21 MPa, front plate 51 MPa, bottom plate 144 MPa (allow 323); fatigue 0.07. Thicknesses are set by machining and slewing ring flatness, not strength: FE stiffness check of the ring seat before release.
5. Welds ISO 5817 B, 100 % MT; stress relieve; machine front and bottom plates in one set-up on a 5° fixture: front face square to the yaw flange within 0.05/400 at 5.00° ± 0.05°; ring seat flatness 0.05.

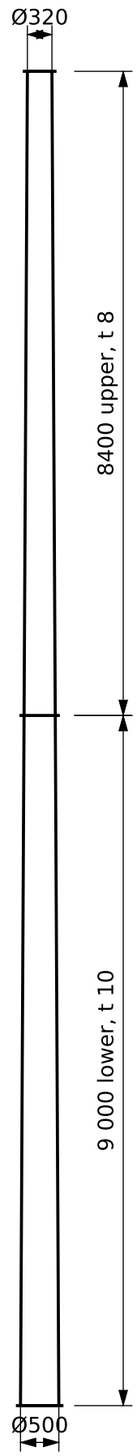
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

BEDPLATE WELDMENT 6 kW (LIGHT)

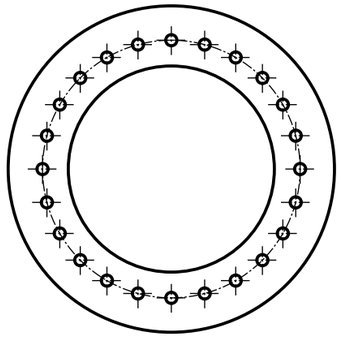
DRAWING NO.	REV / DATE	SCALE (A3)
NS6-300	A · 2026-11-06	1:5
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
S355J2+N plates, stress relieved, machined	127 kg	Claude / not checked

ELEVATION

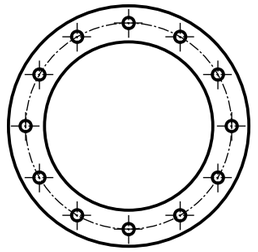


adapter NY-400 on top; rotor centre 18 000

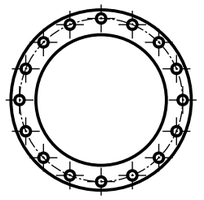
FLANGES (plan)



BASE FLANGE 40, 24 × M24 PCD 600



FLANGE 30 at 9 m, 12 × M24 PCD 480



TOP FLANGE 30, 16 × M20 PCD 380

NOTES

1. Doc 09 design (tilt-up on a base hinge, removable gin pole 0.33 H, no crane) at the doc 30 S6 size; door-free base: cables leave through the base hinge frame into the cabinet.
2. Longitudinal seam welds full penetration, staggered 90° between cans; flange welds full penetration, ISO 5817 B, 100 % UT; flanges machined flat after welding (0.2 mm).
3. Climbing: none (service by lowering the tower). Lightning: tower is the down conductor, 2 earthing lugs at the base.
4. Tower 1P exclusion zone 47.5–57.4 rpm in the controller (doc 31).

Item	Value
Shell	Ø500 × 10 → Ø320 × 8, linear taper, D/t ≤ 51
Sections	2: 9.0 m + 8.4 m, flanged
First frequency	0.87 Hz (soft footing, ice 0.81 Hz)
Governing	parked V_e50, use 0.97
Fatigue	0.06 base, 0.02 at 9 m (doc 31 spectra)
Tilt-up	base moment 405 kNm, 6 m gin pole, winch 68 kN
Base flange	24 × M24 10.9, F_bolt 222.9 kN / 254.2 kN

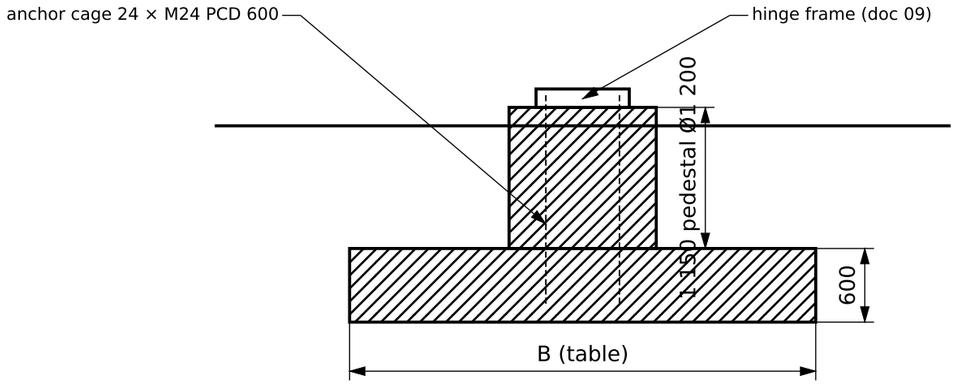
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

TOWER 18 m · TILT-UP MONOPOLE

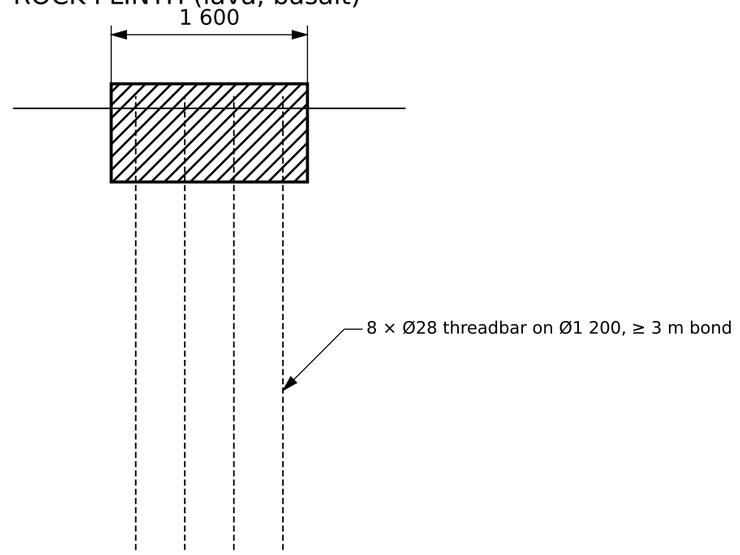
DRAWING NO.	REV / DATE	SCALE (A3)
NS6-400	A · 2026-11-06	1:50 / 1:10
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
S355J2+N, rolled and seam welded; hot-dip galvanized	1824 kg	Claude / not checked

SPREAD FOOTING, SECTION



Ground	B m	Concrete m³	Bearing kPa	q_rd kPa
firm soil (dense gravel, moraine)	3.8	9.5	166.5	300.0
soft volcanic soil (loose ash, pumice sand)	4.2	11.4	95.9	100.0
frost-susceptible silt (thawed)	3.9	10.0	138.2	150.0
rock (anchored plinth)	1.6	2.0	–	anchors 249 kN

ROCK PLINTH (lava, basalt)



NOTES

1. Base 1.6 m below ground (frost depth). Design for the parked V_e50 base moment 617 kNm (doc 30 §6); characteristic eccentricity ≤ B/4, operation ≤ B/6.
2. Concrete C30/37 XF4 with air entrainment, cover 50 mm; rebar mesh top and bottom Ø16/150 both ways (to be detailed with the local engineer).
3. Rock: drill Ø76, grout cement 100 %, proof-load each anchor to 1.25 × design before tensioning; plinth cast on cleaned rock with 20 mm shear keys.
4. Soil class must be confirmed by a site inspection; soft volcanic ash needs a plate-load test.

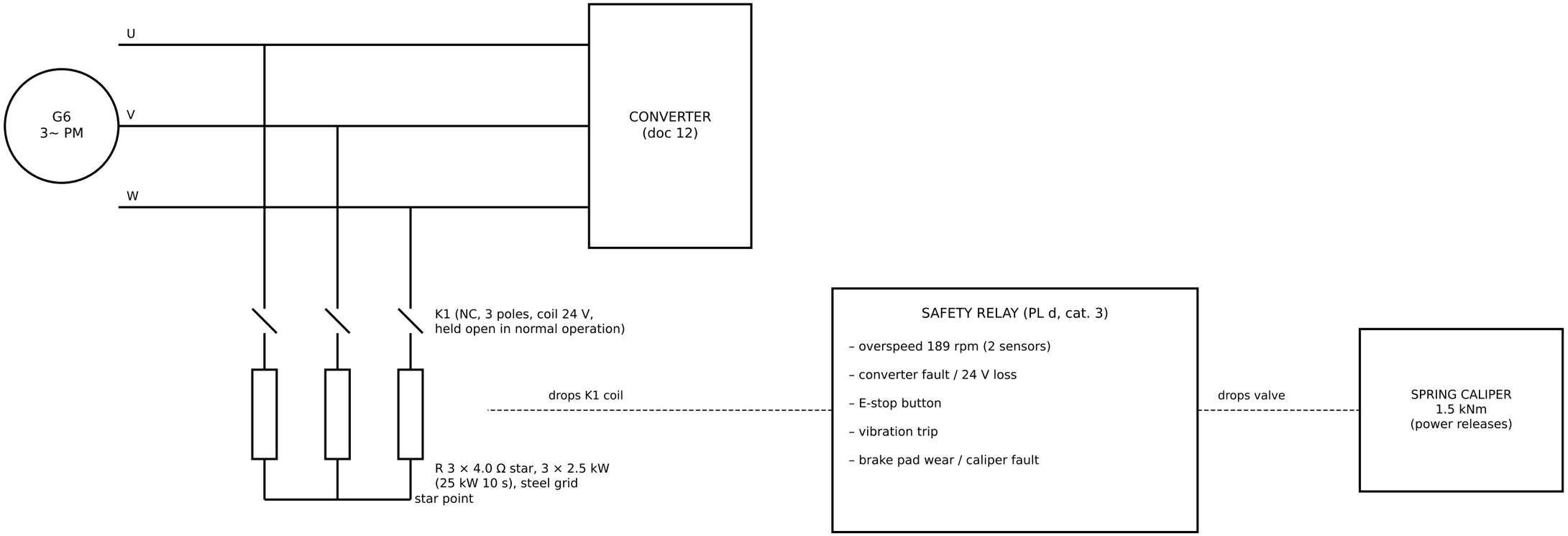
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

FOUNDATION · SPREAD FOOTING AND ROCK PLINTH

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-500	A · 2026-11-06	1:50
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
C30/37 XF4 air-entrained; B500C rebar	9.5–11.4 m³ (soil), 2.0 m³ (rock)	Claude / not checked

SINGLE-LINE: BOTH STOPPING CHANNELS DROP TOGETHER ON A TRIP



NOTES

1. Doc 29 / 31: K1 is normally closed. While the safety relay is healthy it holds K1 open; any trip or loss of 24 V closes K1 and drops the brake at the same time. Either channel alone keeps the rotor below the trip speed (doc 31: brake alone stops in ≤ 2.4 s, peak 173 rpm; resistor alone slows to a 3–15 rpm crawl).
2. Resistor current 43 A peak at trip speed with hot magnets; energy of one stop ≈ 120 kJ. Grid resistors in a ventilated 1.4301 box (600 × 400 × 250) on the tower base cabinet wall, not in the nacelle.
3. K1: 3-pole contactor with mechanically linked mirror contact (EN 60947-4-1 Annex F) for monitoring; AC-3 rating ≥ 63 A at 690 V.
4. Brake fault alarm: with the brake failed the rotor only crawls, so a failed or worn brake keeps the turbine parked and yawed out until repaired (doc 31 §6).
5. Spring pack stays 1.5 kNm (Martin, 2026-11-06). Parked yaw-lost 50-year peak aero torque 1.45 kNm; if exceeded the rotor creeps slowly against the brake and the closed K1 (doc 32 §1). Use the lock pin NP1-320 for long unattended outages.

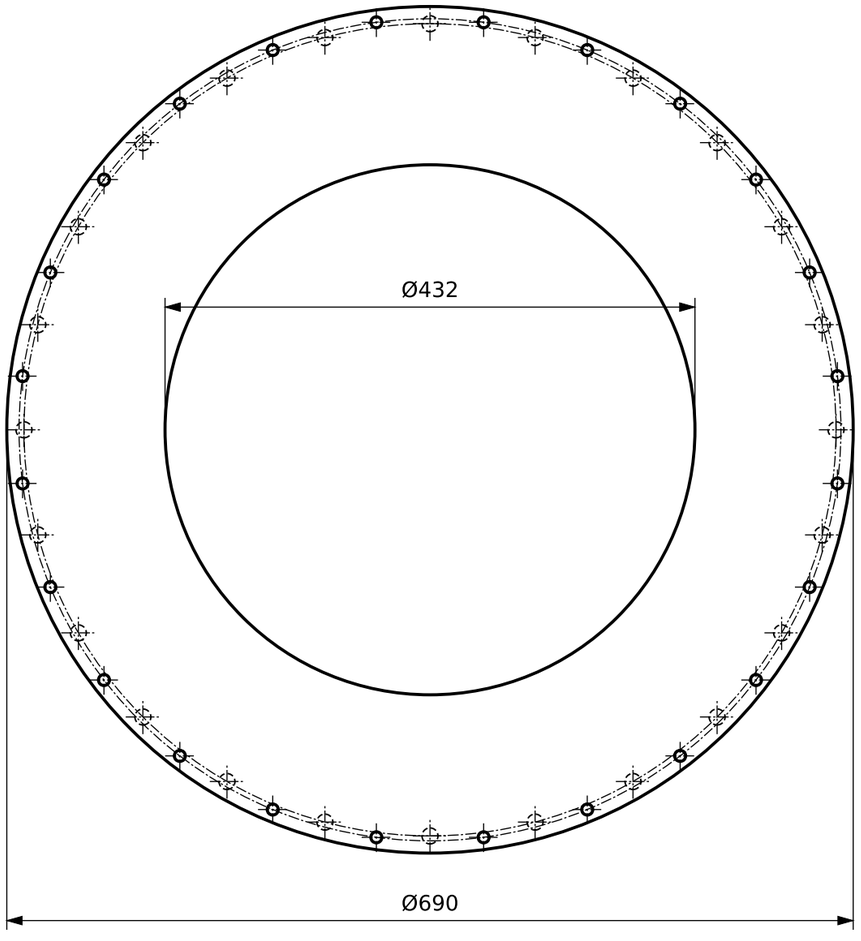
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

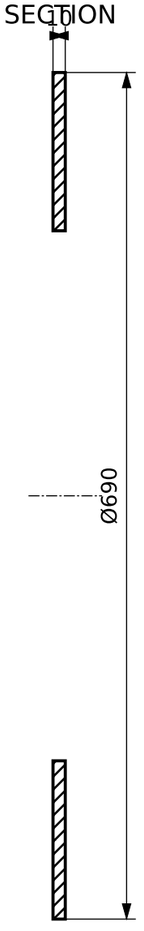
BRAKING RESISTOR BOX AND SAFETY CHAIN

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-600	A · 2026-11-06	schematic / 1:5
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
Box 1.4301, IP54; resistors steel-grid	≈ 18 kg	Claude / not checked

VIEW FROM THE REAR



SECTION



NOTES

1. Replaces the Rev C rear cover with 24 fan blades (doc 23/29): G6 runs at half load with no forced cooling; hot spot 86 °C at 1.2 × rated torque without fins, ribs or blades (doc 29).
2. 24 × M8 on PCD 670 to the drum rear ring; seal carrier as Rev C (NP1-212 inner ring).
3. Rear face runs 5 mm from the bedplate front plate outside the Ø410 pad (as Rev B).

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

G6 PLAIN REAR COVER (NO FAN BLADES)

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-610	A · 2026-11-06	1:5
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
EN AW-5083-H111, 10 mm plate	≈ 7 kg	Claude / not checked